

Debasish Ray Mohapatra

PhD Candidate | University of British Columbia

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EDUCATION

University of British Columbia PhD in Electrical and Computer Engineering Advisor: Dr. Sidney Fels, P.Eng.	Vancouver, Canada ongoing
University of British Columbia MAsc in Electrical and Computer Engineering Advisor: Dr. Sidney Fels, P.Eng.	Vancouver, Canada Apr 2021
Siksha 'O' Anusandhan University B.Tech. in Electronics and Communication Engineering	Bhubaneswar, India Aug 2013

EMPLOYMENT

Tata Consultancy Services 🌐 Software Test Engineer Project: Functional and regression testing of the front-end (Web app) and back-end (ETL system) applications.	Chennai, India London, UK Jan 2014 - Mar 2017
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RESEARCH EXPERIENCE

Smule, Inc. 🌐 Student Researcher (Remote) Project: Developing a 3D acoustic wave solver in CUDA and assist with the physics-based modeling of the human vocalization system. Advisor: Yongyi Zang, Dr. Perry R. Cook, Dr. Randal Leistikow	Seattle, USA Aug 2025 - Jul 2026
VocalTractLab 🌐 Visiting Research Scholar Project: A comparative analysis of vocal tract centreline determination algorithms. Advisor: Dr. Peter Birkholz	Dresden, Germany Jun 2022 - Aug 2022
Human Communication Technologies Lab 🌐 Graduate Research Assistant Project: Vocal tract acoustic modeling. Advisor: Dr. Sidney Fels, P.Eng.	Vancouver, Canada Sep 2018 - Present

TEACHING EXPERIENCE

University of British Columbia Teaching Assistant Courses: Human-Computer interfaces in engineering design, CPEN 441 Introduction to computation in engineering design, APSC 160 Introduction to microcomputers, CPEN 211	Vancouver, Canada
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Courses:

Computational thinking, CPSC 100

Basic algorithms and data structures, CPSC 221

AWARDS, FELLOWSHIPS & GRANTS

PhD CoLab Grant (8000 CAD) University of British Columbia <i>Research grant for collaborative and interdisciplinary scholarly work.</i>	2024
Interspeech Student Travel Grant (600 EUR) International Speech Communication Association (ISCA) and Interspeech <i>Travel grant to attend Interspeech 2022 at Incheon, South Korea</i>	2022
Language Sciences Trainee Travel Fund (3000 CAD) University of British Columbia <i>Travel fund for a short-term visit to VocalTractLab at TU Dresden, Germany</i>	2022
Graduate Student Travel Award (500 CAD) University of British Columbia	2022
Go Global Self-Directed Research Award (1500 CAD) University of British Columbia <i>Travel fund for a short-term visit to VocalTractLab at TU Dresden, Germany</i>	2022
President's Academic Excellence Initiative PhD Award University of British Columbia	2021-Present
International Tuition Award University of British Columbia	Sep 2018 - Dec 2023
Certification of Appreciation for outstanding contribution Tata Consultancy Services	2015

OPEN SOURCE

2.5D Vocal Tract Model [🔗] The 2.5D Finite-Difference Time-Domain (FDTD) acoustic wave solver to simulate wave propagation in static mirror-symmetric wave guides.	2026
3D Vocal Tract Model [🔗] A 3D Finite-Difference Time-Domain (FDTD) acoustic wave solver to simulate sound propagation in static vocal tract geometries.	2022

JOURNALS

- [1] **A 2.5-Dimensional Acoustic Wave Solver: Modeling simplified vocal tract geometries with reduced computational load**
Debasish Ray Mohapatra, Victor Zappi, Sidney Fels
The Journal of the Acoustical Society of America, Vol. 159(4), pp. 3515-3532 **[JASA '26]**

PEER-REVIEWED CONFERENCE PROCEEDINGS

- [10] **Influence of Vocal Tract Curvature on Speech Acoustics: A Three-Dimensional FEM Analysis**
Debasish Ray Mohapatra, Sidney Fels
Proceedings of Interspeech, pp. XX-XX (Accepted) [Interspeech '26]
- [9] **Physics-Informed Neural Operator for Speech Production Analysis**
Kazuya Yokota, Xinmeng Luan, Debasish Ray Mohapatra, Gary Scavone, Sidney Fels
Proceedings of Interspeech, pp. XX-XX (Accepted) [Interspeech '26]
- [8] **Beyond Direct Geometry: Spring-Mass control of tongue articulation for vocal synthesis**
Debasish Ray Mohapatra, Ziyi Xia, Sidney Fels
New Interfaces for Musical Expression [NIME '26]
- [7] **2D Immersed Boundary Method in vocal tract acoustics: An Eulerian–Lagrangian model for simulation of diphthongs**
Rongshuai Wu, Debasish Ray Mohapatra, Sidney Fels
Proceedings of Interspeech, pp. 973-977 [Interspeech '25]
- [6] **Modeling vocal tract like acoustic tubes using the immersed boundary method**
Rongshuai Wu, Debasish Ray Mohapatra, Sidney Fels
Proceedings of Interspeech, pp. 3415-3419 [Interspeech '24]
- [5] **2.5D Vocal Tract Modeling: Bridging low-dimensional efficiency with 3D accuracy**
Debasish Ray Mohapatra, Victor Zappi, Sidney Fels
Proceedings of Interspeech, pp. 17-21 [Interspeech '24]
- [4] **Three-dimensional finite-difference time-domain acoustic analysis of simplified vocal tract shapes**
Debasish Ray Mohapatra, Mario Fleischer, Victor Zappi, Peter Birkholz, Sidney Fels
Proceedings of Interspeech, pp. 764-768 [Interspeech '23]
- [3] **Vocal tract area function extraction using ultrasound for articulatory speech synthesis**
Debasish Ray Mohapatra, Prमित Saha, Yadong Liu, Bryan Gick, Sidney Fels
Speech Synthesis Workshop, pp. 90-95 [SSW '21]
- [2] **A comparative study of two-dimensional vocal tract acoustic modeling based on Finite-Difference Time-Domain methods**
Debasish Ray Mohapatra, Victor Zappi, Sidney Fels
International Seminar on Speech Production, pp. 154-157 [ISSP '20]
- [1] **An extended two-dimensional vocal tract model for fast acoustic simulation of single-axis symmetric three-dimensional tubes**
Debasish Ray Mohapatra, Victor Zappi, Sidney Fels
Proceedings of Interspeech, pp. 3760-64 [Interspeech '19]

LIGHTLY-REVIEWED ABSTRACTS AND PREPRINTS

- [2] **Sound-Stream: Towards real-time gesture controlled articulatory sound synthesis**
Prमित Saha, Debasish Ray Mohapatra, Praneeth Srungarapu, Sidney Fels
Proceedings of Acoustic Week in Canada, pp. 58-59 [AWC '18]
- [1] **Limitations of source-filter coupling in phonation**
Debasish Ray Mohapatra, Sidney Fels
Proceedings of Acoustic Week in Canada, pp. 60-61 [AWC '18]

ACADEMIC SERVICES

Journal Reviewing

Archives of Acoustics [2022(1)]
Acta Acustica [2024(1)]
Journal of the Acoustical Society of America [2025(1)]
JASA Express Letter [2025(1)]

Conference Reviewing

Interspeech [2025(3), 2026(4)]

STUDENTS MENTORED

[1] **Rongshuai Wu, M.A.Sc.** 
University of British Columbia, Canada
Thesis: *Dynamic vocal tract acoustic modeling using the immersed boundary method*

SERVICE & VOLUNTEERING

STEM Outreach: Introductory AI Tutor — Mother’s Public School (2026) Mentored middle-school students on foundational AI concepts, introducing basic machine learning ideas and real-world AI applications.	Bhubaneswar, India
Human Communication Technologies Lab Ambassador (2020-21) Worked as a lab representative for the HCT lab at UBC	Vancouver, Canada